Munawar Kazmi

munawarsaeedkazmi1@gmail.com | munawarkazmi.com | github.com/munawarkazmi | linkedin.com/in/munawar-kazmi

Profile

Robotics and AI engineer, MSc (Commendation) in AI and Robotics, working on AI reliability and human-robot interaction. Author of plan-failure-bench, a machine-checkable benchmark of how LLM planners fail (DOI 10.5281/zenodo.21756817); HRI research on the iCub humanoid, under review at MDPI Robotics; ships safety-critical systems from simulation to deployed hardware, with every published number reproducible from public, CI-tested repositories.

Featured Research

plan-failure-bench github.com/munawarkazmi/plan-failure-bench

Sole author. A machine-checkable benchmark of how LLM planners fail. Python, PDDL, differential testing, CI. Working paper on Zenodo.

- Designed 60 proof-carrying instructions over two symbolic environments, six planted trap families plus valid controls; every ground-truth label is a mechanical proof re-verified by 548 CI tests on each commit.
- Judge-free scoring: a deterministic checker, differentially tested against an independent PDDL toolchain, assigns one verdict per response; detection is never reported without its paired false positive count.
- Ran an 18-run grid over four models in plain and semantically obfuscated conditions; the versioned obfuscation caught and retired two artefacts of its own first token scheme.

Trust-calibrated HRI on the iCub humanoid, MSc thesis University of Hertfordshire

Isaac Sim digital twin, ROS2, physical deployment. Supervised by Prof. Patrick Holthaus.

- Built a sim-to-real pipeline transferring verbal and non-verbal intent communication to physical hardware; measured a significant clarity effect (4.9 vs 5.8 on a 7-point scale, $p = 0.017$, $d = 0.58$) in structured user trials.

Selected Projects

llm-nav-shield: detect, recover, or halt github.com/munawarkazmi/llm-nav-shield

- Neurosymbolic navigation shield composing two verified cores (pinned submodules): a deterministic trajectory verifier that caught 35/35 unsafe qwen2.5-7B and 32/32 unsafe Llama-3.3-70B proposals with zero misses, and a provably-correct classical planner for recovery.
- On the committed evaluation: 38/38 unsafe proposals recovered to verified-safe paths, 10/10 correct halts when no safe path exists, zero unsafe trajectories forwarded; every number replayed in CI.

ROS2 dynamic path planning: A* and D* Lite for Nav2 github.com/munawarkazmi/ros2-dynamic-path-planning

- ROS-free C++20 planning core with Nav2 plugin adapters; in the seeded blocked-route benchmark (200 trials, 1,493 replan events) D* Lite replans 4.3x faster on average and 11x on the median, with both planners' path costs matching on all 1,716 measurements and optimality fuzz-validated against Dijkstra over 185,237 replans.

Experience

Full-Stack Software Engineer (Contract), Safina International Multan, Pakistan; Jul 2026 - present

- Sole engineer of the institute's live management platform (portal.safinainternational.com): seven user roles, prorated billing, payroll, and an append-only financial ledger enforced via PostgreSQL row-level security.

Visiting Lecturer, AI and Robotics, Pak-Turk Maarif Schools Multan, Pakistan; Aug 2025 - present

- Designed and teach an ML curriculum for 30+ students; student teams reached 92% model accuracy on custom classification tasks.

Embedded Systems Engineer (Contract), Muxtronics Multan, Pakistan; Jan 2025 - May 2025

- Deterministic C++ firmware for real-time ESP32 image processing and a LoRa mesh with sub-3-second alert latency at 100 m, shipped in a safety-critical product.

Robotics Integration Specialist, Robot House, University of Hertfordshire Hatfield, UK; Jan 2024 - May 2024

- Gesture recognition and motion planning pipeline (Unity, C++) for the iCub; deployed PyTorch communication models into its real-time control loop.

Research Associate, University of Hertfordshire Hatfield, UK; May 2023 - Dec 2023

- Optimised TensorFlow LLM inference for the Unitree Go1 quadruped, cutting latency 29% across 200 trials.

Education

MSc Artificial Intelligence and Robotics, Commendation, University of Hertfordshire 2023 - 2024
BEng Mechatronics Engineering, National University of Sciences and Technology (NUST) 2018 - 2022

Publications

Kazmi, M. (2026). plan-failure-bench: A Machine-Checkable Benchmark of How Language Model Planners Fail. Working paper, Zenodo. DOI 10.5281/zenodo.21756817.
Kazmi, M. (2025). Enhancing Human-Robot Companionship: Comparing Verbal and Non-Verbal Communication Cues in Humanoid Robots. Manuscript under review, MDPI Robotics.

Technical Skills

C++20, Python, ROS2 (Nav2, plugins); differential and property-based testing, CI/CD, Docker, Git; PyTorch, TensorFlow, TensorRT, LLM evaluation and integration; Isaac Sim, MuJoCo, Gazebo, Unity; ESP32, LoRa mesh, bare-metal C/C++, Jetson; React, TypeScript, PostgreSQL (row-level security)